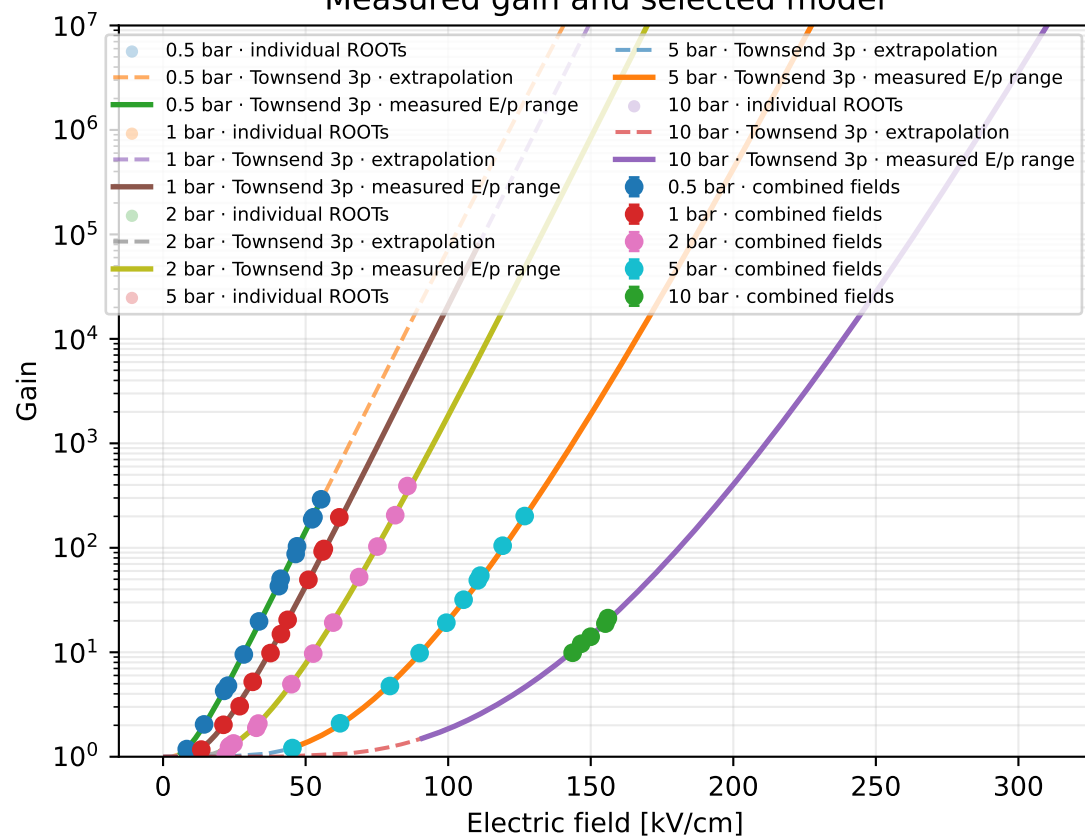
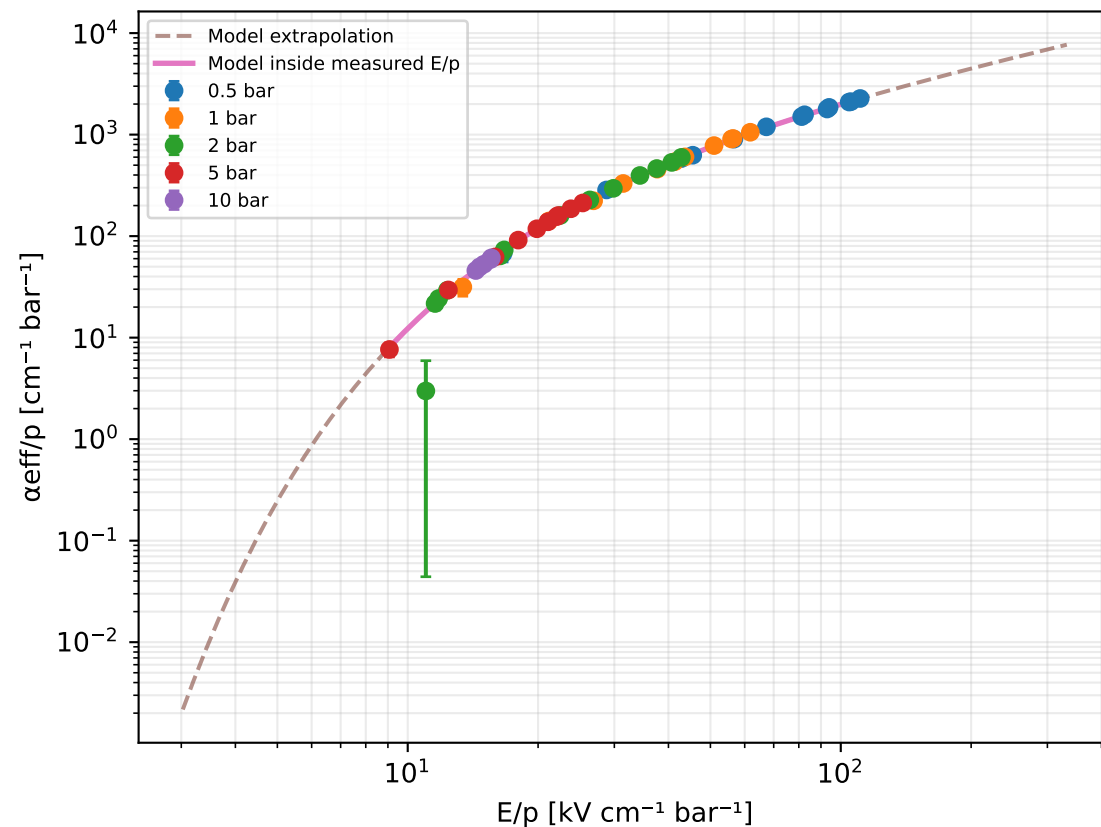


ArCF4 · Ar 100 % / CF4 0 % · gap 0.05 mm

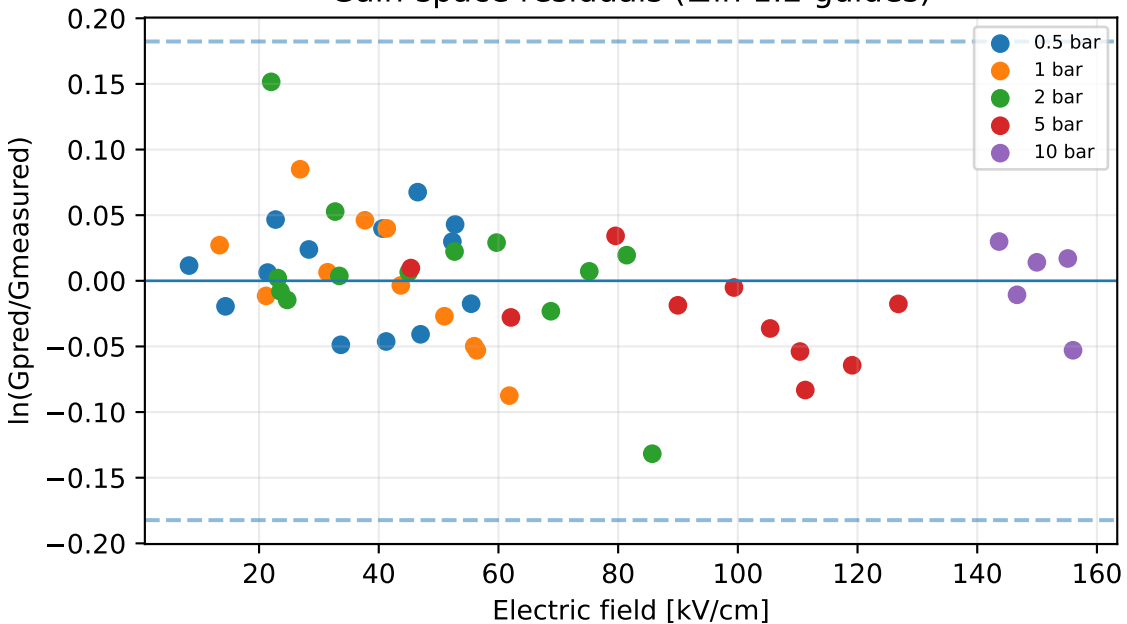
Measured gain and selected model



Reduced Townsend fit

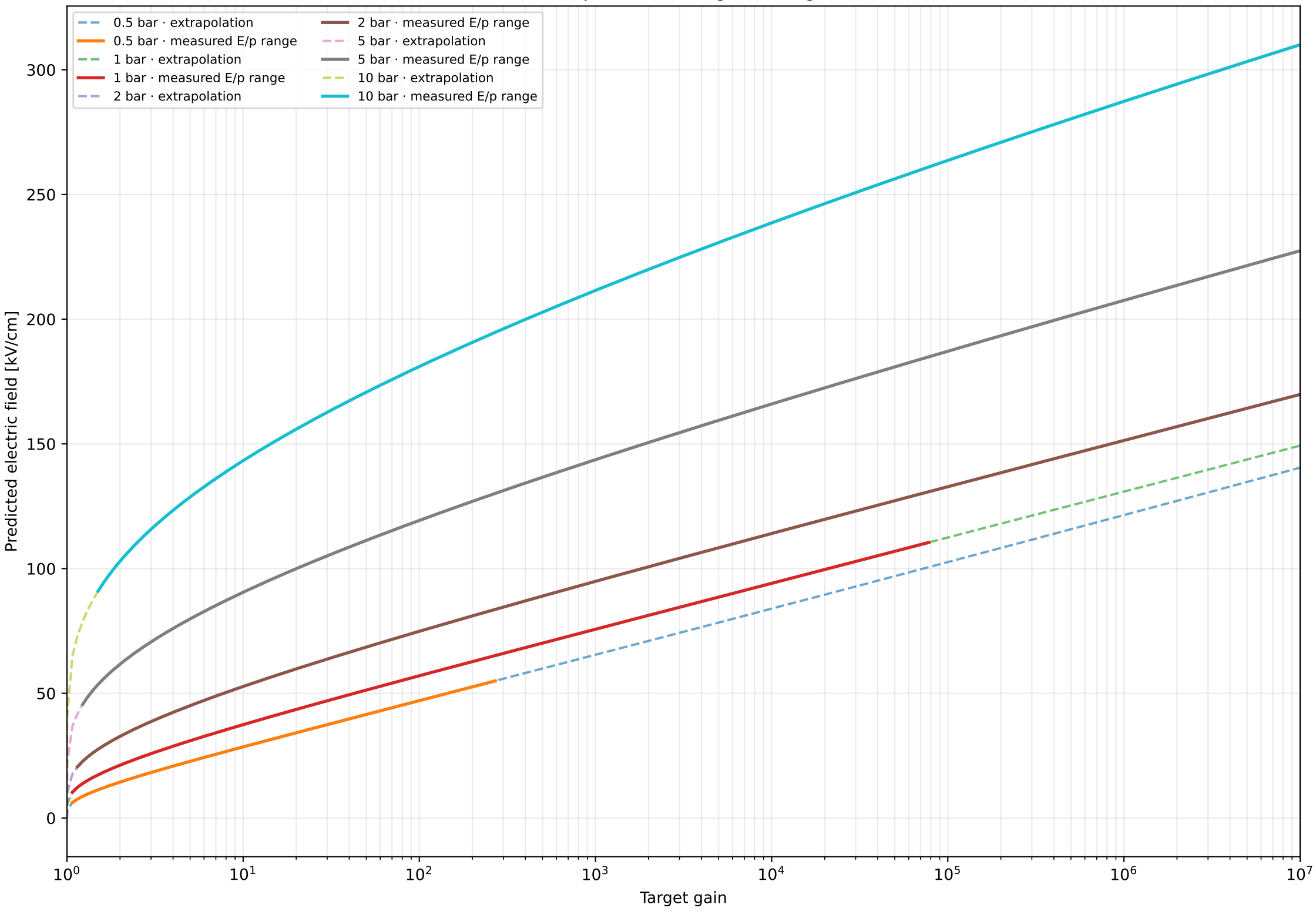


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 67
 Unique physical points: 52
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 3p
 Status: accepted
 Training RMSE: $\times 1.287$
 Cross-validation median: $\times 1.030$
 Cross-validation worst: $\times 1.164$
 Reduced-field range: 9.073–110.9 kV cm $^{-1}$ bar $^{-1}$
 $A=37.276$, $B=32.617$, $m=0.93456$, $n=1$
 Covariance condition: 1.13e+04

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
0.5	8.30433	16.6087	1.184211	0.0354	33.8153	152	1
0.5	14.3923	28.7845	2.039548	0.0612	142.546	354	1
0.5	21.4253	42.8506	4.273148	0.128	290.47	432	1
0.5	22.7576	45.5152	4.791667	0.144	313.376	456	1
0.5	28.3237	56.6474	9.536585	0.286	451.027	369	1
0.5	33.6632	67.3263	19.7486	0.592	596.617	358	1
0.5	40.6193	81.2387	42.95441	1.29	752.028	329	1
0.5	41.2347	82.4693	50.55733	1.52	784.622	375	1
0.5	46.5022	93.0044	87.2792	2.62	893.822	351	1
0.5	46.9855	93.9709	103.3383	3.1	927.602	269	1
0.5	52.2965	104.593	187.47	5.64	1046.72	300	1
0.5	52.757	105.514	196.0367	6.48	1055.66	300	1
0.5	55.4343	110.869	291.336	8.72	1134.9	253	1
1	13.4043	13.4043	1.17094	0.035	31.5614	117	1
1	21.1748	21.1748	2.021978	0.0606	140.815	364	1
1	26.8704	26.8704	3.047714	0.0913	222.878	503	1
1	31.4447	31.4447	5.212066	0.156	330.195	547	1
1	37.6881	37.6881	9.826786	0.294	457.022	560	1
1	41.3391	41.3391	14.93346	0.448	540.721	511	1
1	43.6849	43.6849	20.43605	0.612	603.46	516	1
1	50.9723	50.9723	49.42236	1.48	780.081	483	1
1	55.9562	55.9562	92.19667	3.5	904.785	300	1
1	56.3881	56.3881	97.46347	2.92	915.896	438	1
1	61.8147	61.8147	195.7467	6.57	1055.36	300	1
2	22.0233	11.0117	1.030303	0.0303	5.97059	33	1
2	23.1334	11.5667	1.242236	0.0372	43.3826	161	1
2	23.5748	11.7874	1.275	0.0381	48.5892	200	1
2	24.701	12.3505	1.340426	0.0402	58.5974	235	1
2	32.7166	16.3583	1.894089	0.0568	127.748	406	1
2	33.4128	16.7064	2.078029	0.0622	146.284	487	1

p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
2	45.0086	22.5043	4.949917	0.148	319.874	599	1
2	52.6645	26.3322	9.716561	0.291	454.766	628	1
2	59.6864	29.8432	19.23774	0.577	591.375	673	1
2	68.7588	34.3794	52.45918	1.57	792.007	588	1
2	75.1761	37.5881	102.8217	3.24	926.599	600	1
2	81.4227	40.7114	205.2467	8.67	1064.84	300	1
2	85.7012	42.8506	390.0332	11.7	1193.25	572	1
5	45.3658	9.07317	1.210811	0.0363	38.258	185	1
5	62.0852	12.417	2.090559	0.0626	147.486	519	1
5	79.5709	15.9142	4.75583	0.143	311.874	729	1
5	89.982	17.9964	9.823232	0.295	456.95	792	1
5	99.3528	19.8706	19.19125	0.575	590.891	800	1
5	105.404	21.0807	31.85452	0.954	692.236	818	1
5	110.385	22.077	48.83353	1.46	777.683	847	1
5	111.269	22.2538	54.16403	1.62	798.403	823	1
5	119.104	23.8208	104.7367	3.67	930.29	600	1
5	126.825	25.365	201.32	10.1	1060.98	300	1
10	143.651	14.3651	9.895506	0.297	458.416	890	1
10	146.649	14.6649	12.06556	0.0906	498.071	13850	16
10	149.965	14.9965	14.09214	0.422	529.123	814	1
10	155.13	15.513	18.81001	0.564	586.878	879	1
10	156.009	15.6009	21.22776	0.637	611.062	843	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-127	×1.33	×6.73	no	large cross validation outlier
Townsend 3p	3	-137	×1.03	×1.16	yes	—
Townsend 4p	4	-134	×1.03	×1.16	yes	—